

# Actuator-Constrained Reinforcement Learning for High-Speed Quadrupedal Locomotion

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**Abstract**—This paper presents a method for achieving high-speed running of a quadruped robot by considering the actuator torque-speed operating region in reinforcement learning. The physical properties and constraints of the actuator are included in the training process to reduce state transitions that are infeasible in the real world due to motor torque-speed limitations. The gait reward is designed to distribute motor torque evenly across all legs, contributing to more balanced power usage and mitigating performance bottlenecks due to single-motor saturation. Additionally, we designed a lightweight foot to enhance the robot’s agility. We observed that applying the motor operating region as a constraint helps the policy network avoid infeasible areas during sampling. With the trained policy, KAIST Hound, a 45 kg quadruped robot, can run up to 6.5 m/s, which is the fastest speed among electric motor-based quadruped robots.

## I. INTRODUCTION

Legged robots can leverage reinforcement learning (RL) techniques to enhance their ability to execute challenging tasks. The successful implementation of reinforcement learning algorithms has shown that robots can navigate through challenging environments, which include uneven, unstructured, and deformable terrains [1]–[4].

These approaches are based on simulations where robots can collect data. By repeatedly interacting with their simulated environments, the agents can gradually improve their performance.

However, the simulation-to-reality (sim-to-real) gap remains one big challenge that needs to be addressed. This issue comes from the differences between the conditions in simulations and the unmodeled factors of real-world environments. Because of this gap, the real robot behavior and performance can change when transitioning from a simulation to a real world. To mitigate this gap, previous studies have employed domain randomization techniques. These techniques introduce a range of variability within simulation parameters, thereby enhancing the adaptability of reinforcement learning agents to the inherent unpredictability of real-world scenarios. There are attempts to randomize the training environment, such as terrain, so that the robot can walk robustly on various terrains [1], [2]. To overcome the inaccurate system identification, such as torque constant or mass of the robot, noise is added to the robot parameters [3], [5]. Hwangbo et al. [6] mapped torque commands from a trained network to torques measured in the real world through a network to predict the behavior of

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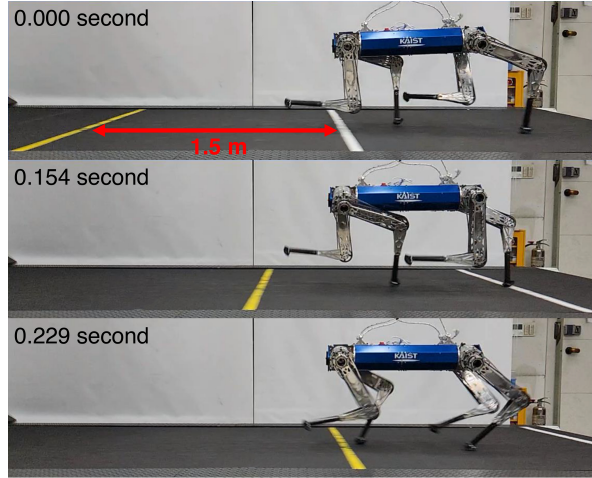


Fig. 1. KAIST Hound crosses a 1.5-meter distance between two lines on the treadmill in 0.229 seconds with an average speed 6.5 m/s. The snapshots are obtained from a video recorded with a rate of 240 frames per second.

actuators such as series elastic actuators, which is difficult to model in simulation. These difficulties include non-linearity of the actuator and lack of information from the manufacturer.

Besides the aforementioned research related to stable locomotion, there have been efforts to achieve highly dynamic motion through reinforcement learning. In challenging motions such as high-speed running, the motor output is used close to the boundary of its capability [7]. Therefore, studies have been conducted that reflect the characteristics of the motor in the simulator to find feasible trajectories in the real world for modeled acrobatic behaviors [8].

To integrate motor modeling into reinforcement learning simulators, we imposed set of linear inequality constraints representing feasible motor torque and angular velocity regions during training. This effectively clips the infeasible area in the real world to a feasible range for torque commands. These constraints correspond to the motor’s operating range, readily available from its specification sheet.

By clipping motor output in the simulation for commands that would be infeasible, we prevented state transitions in the simulation that cannot be realized in the real world. This approach allows the robot to adapt torque commands that exceed the maximum capacity of the motor, even in the real world. We also observed that the proposed method helps the network avoid sampling the infeasible range of motor outputs during the learning process.

Our methodology was demonstrated using KAIST Hound [9], which is a quadruped robot designed for high-speed running with optimized gear ratios. Using the proposed method, KAIST Hound, a 45 kg robot, can robustly run at speeds up

# 面向高速四足运动的执行器约束强化学习

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**摘要**—本文提出了一种在强化学习中考虑执行器转矩-转速工作区域以实现四足机器人高速奔跑的方法。训练过程中纳入了执行器的物理特性和约束，以减少因电机转矩-转速限制而在真实世界中不可行的状态转换。步态奖励的设计旨在将电机转矩均匀分配至所有腿部，有助于更均衡的功率使用，并缓解因单电机饱和导致的性能瓶颈。此外，我们设计了轻量化足部以增强机器人的敏捷性。我们观察到，将电机工作区域作为约束有助于策略网络在采样过程中避开不可行区域。凭借训练后的策略，KAIST Hound（一台45 kg的四足机器人）能够以高达6.5 m/s的速度奔跑，这是基于电机驱动的四足机器人所达到的最快速度。

## I. 引言

腿式机器人可以利用强化学习（RL）技术来增强其执行具有挑战性任务的能力。强化学习算法的成功实施表明，机器人能够穿越具有挑战性的环境，包括不平坦、非结构化和可变形的地形 [1]–[4]。

这些方法基于仿真，机器人可以在仿真中收集数据。通过与仿真环境反复交互，智能体可以逐步提高其性能。

然而，仿真到现实（sim-to-real）的差距仍然是一个需要解决的大挑战。这个问题源于仿真环境条件与现实世界环境中未建模因素之间的差异。由于这种差距，真实机器人的行为和性能在从仿真过渡到现实世界时可能会发生变化。为了缩小这一差距，以往的研究采用了域随机化技术。这些技术在仿真参数中引入一系列可变性，从而增强强化学习智能体对现实世界场景固有不可预测性的适应能力。有研究尝试随机化训练环境（如地形），以使机器人能够在各种地形上稳健行走[1], [2]。为了克服不准确的系统辨识（如转矩常数或机器人质量），会在机器人参数中添加噪声 [3], [5]。Hwangbo等人 [6]通过一个网络将训练好的网络输出的力矩指令映射到真实世界中测得的转矩，以预测

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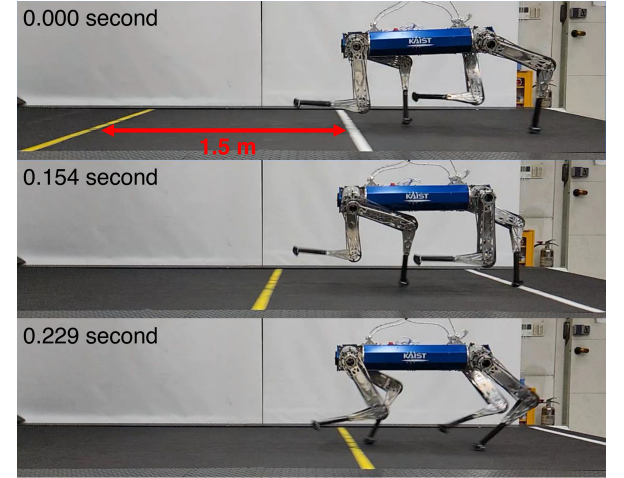


图1. KAIST Hound在跑步机上以6.5米/秒的平均速度在0.229秒内跨越两条线之间1.5米的距离。快照取自以每秒240帧速率录制的视频。

诸如串联弹性执行器之类的执行器，在仿真中难以建模。这些困难包括执行器的非线性以及制造商提供的信息不足。

除了上述与稳定运动相关的研究外，人们还致力于通过强化学习实现高度动态的运动。在高速奔跑等具有挑战性的运动中，电机输出被使用到接近其能力边界 [7]。因此，已有研究在模拟器中反映电机的特性，以在真实世界中为建模的杂技行为寻找可行轨迹 [8]。

为了将电机建模整合到强化学习模拟器中，我们在训练过程中施加了一组线性不等式约束，表示可行的电机转矩和角速度区域。这有效地将真实世界中不可行的区域裁剪为力矩指令的可行范围。这些约束对应于电机的工作范围，可从其规格说明书中直接获取。

通过在仿真中对不可行的指令进行电机输出裁剪，我们避免了仿真中无法在真实世界中实现的状态转换。这种方法使机器人即使在真实世界中也能适应超过电机最大能力的力矩指令。我们还观察到，所提方法有助于网络在学习过程中避免采样不可行的电机输出范围。

我们的方法通过KAIST Hound[9], 进行了验证，这是一款专为高速奔跑而设计、配备优化传动比的四足机器人。采用所提方法后，重达45公斤的KAIST Hound能够稳健地以高达



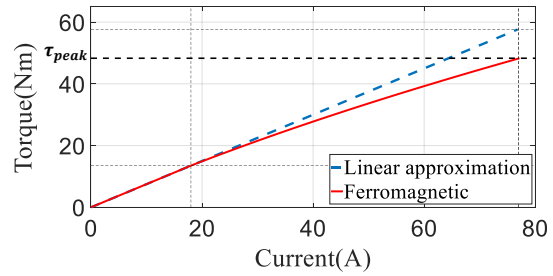


Fig. 2. Motor current versus motor torque of nonlinear ferromagnetic curve and dashed line for linear approximation.

to 6.5 m/s on a treadmill. This marks the fastest running speed recorded for an electric motor-based legged robot, surpassing the previous record of 6.4 m/s set by MIT Cheetah 2 [10]. Additionally, when controlled by reinforcement learning methodology, the highest speed achieved by a quadruped robot was previously reported to be 5 m/s [7].

The remainder of the paper is organized as follows. Section II introduces the motor operating range constraints and overall training process. Section III verifies the proposed method with experimental results in both simulation and the real-world. Section IV concludes this paper.

## II. METHODS

This section introduces our overall framework that enabled high-speed running of quadrupedal robots including policy training using RL and hardware modifications. Our RL framework employs linear inequality constraints to reflect more realistic motor operation regions and introduces a method for obtaining the gearbox matrix essential for the tranformation between motor space and joint space. The paper defines motor space as the input side without the reducer and joint space as the output side after the reducer. Further, we outline a gait reward strategy, and a learning framework focused on torque distribution and bridging the sim-to-real gap. Additionally, the development of a lightweight foot design aimed at enhancing robot agility and operational efficiency is discussed.

### A. Motor operating region (MOR)

The operating region of motors on  $\tau - \omega$  space typically represents a trapezoid-like shape in the first quadrant as shown in Fig. 3 (a), characterizing the motors performance [11], [12]. These geometries come from typical DC motor dynamics, considered the motor as an electrical circuit. For simplicity, assuming that the time rate of current change in a DC motor is negligible. This assumption is the same as when using a dynamometer to collect static data points and then plotting a  $\tau - \omega$  curve [13]. Under quasi-stationary conditions, the equation of DC motor dynamics can be obtained as follows:

$$\begin{aligned} V_{\text{applied}} &= iR + L \frac{di}{dt} + V_{\text{BackEMF}} \\ &\approx iR + V_{\text{BackEMF}} \\ &= \frac{R}{K_t} \tau + \frac{1}{K_v} \omega, \end{aligned} \quad (1)$$

where  $V_{\text{applied}}$  is the voltage applied to counteract the motor's resistance  $R$ ,  $L$  is winding inductance,  $i$  is the motor current,

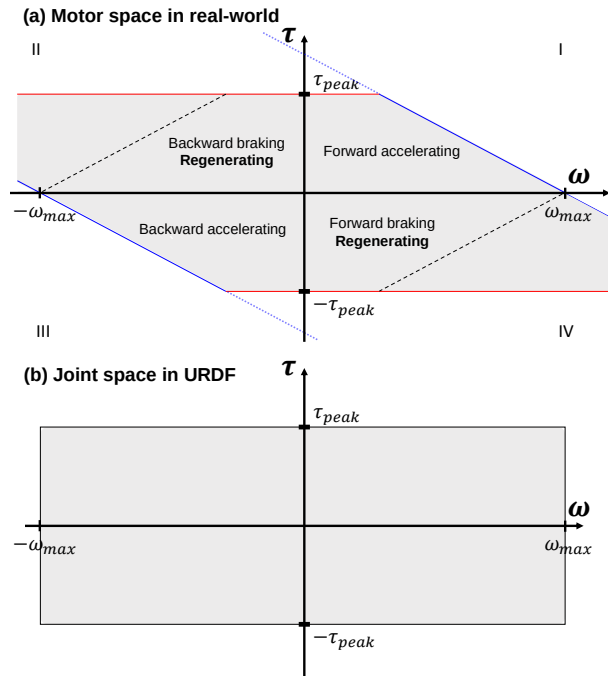


Fig. 3. (a) An operating region of BLDC motor with grey-colored based on the assumption that ignored the time derivative of the current. (b) Torque-velocity constraints defined by the URDF in joint space.

$V_{\text{BackEMF}}$  is the electromotive force produced by the motor due to its rotation,  $K_v$  is the angular velocity constant,  $K_t$  is the torque constant.

Equation (1) describes torque and angular velocity relationship of the motor under the applied voltage. Therefore, the operating region of the motor can be formulated as:

$$-V_{\text{bus}} \leq \frac{R}{K_t} \tau + \frac{1}{K_v} \omega \leq V_{\text{bus}}, \quad (2)$$

where  $V_{\text{bus}}$  is the bus voltage. This relationship is represented by the area between the blue lines in Fig. 3 (a).

For simplicity, some researchers assume that applied current and motor torque are linearly proportional [14], [15]. However, in the case of a motor with a ferromagnetic core, which is widely used for efficiency and stability, the flux saturation of the core causes a nonlinearity, as shown in Fig. 2 [12]. The torque is limited to a peak torque  $\tau_{\text{peak}}$  value, which is defined when the deviation from linearity is 20% [16]. The limit of the motor operating region by  $\tau_{\text{peak}}$  is represented by the area between the red lines in Fig. 3 (a). Note that  $\tau_{\text{peak}}$  is not an absolute value but rather a recommended value that should not be exceeded to prevent the windings from burning out due to high temperature. When plotting this characteristic of a motor over the quadrants, the second and fourth quadrants are where the motor decelerates, and the sign of the work generated by the motor is negative. In physical interpretation, the motor acts as a generator, and the back EMF can contribute to the total voltage. In these quadrants(regenerative braking or generator mode), the operating range can be wider compared to the first and third quadrants, as the motor can potentially generate voltages through back EMF that exceed what the battery alone can provide. The resulting operating range of the motor is represented in Fig. 3 (a).

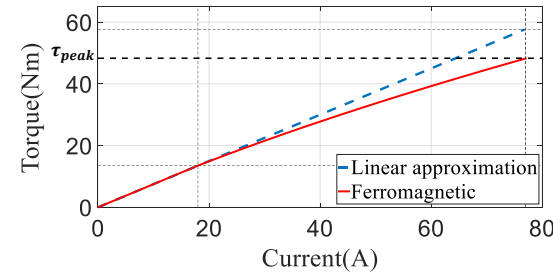


图2. 非线性铁磁曲线下的电机电流与电机转矩关系，以及用于线性近似的虚线。

6.5米/秒的速度在跑步机上奔跑。这标志着基于电动机的腿式机器人所创下的最高跑步速度纪录，超越了此前由MIT猎豹2 [10]保持的6.4米/秒纪录。此外，在强化学习方法的控制下，四足机器人此前报道的最高速度仅为5米/秒 [7]。

本文其余部分组织如下。第II节介绍电机工作范围约束和整体训练过程。第III节通过仿真和现实世界的实验结果验证所提方法。第IV节对本文进行总结。

## II. 方法

本节介绍我们的整体框架，该框架实现了四足机器人的高速奔跑，包括使用强化学习的策略训练和硬件改造。我们的强化学习框架采用线性不等式约束来反映更真实的电机工作区域，并介绍了一种获取减速箱矩阵的方法，该矩阵对于电机空间与关节空间之间的转换至关重要。本文将电机空间定义为不含减速器的输入侧，将关节空间定义为减速器后的输出侧。此外，我们概述了步态奖励策略，以及一个专注于扭矩分配和弥合仿真到现实差距的学习框架。另外，还讨论了旨在增强机器人敏捷性和运行效率的轻量化足部设计的开发。

### A. 电机工作区域 (MOR)

电机在  $\tau - \omega$  空间中的工作区域通常在第一象限呈现梯形形状，如图3 (a) 所示，该区域表征了电机的性能 [11], [12]。这些几何形状源于典型的直流电机动力学，将电机视为一个电路。为简化起见，假设直流电机中电流随时间的变化率可以忽略不计。这一假设与使用测功机采集静态数据点并绘制  $\tau - \omega$  曲线 [13]时的做法相同。在准稳态条件下，直流电机动力学方程可表示如下：

$$\begin{aligned} V_{\text{applied}} &= iR + L \frac{di}{dt} + V_{\text{BackEMF}} \\ &\approx iR + V_{\text{BackEMF}} \\ &= \frac{R}{K_t} \tau + \frac{1}{K_v} \omega, \end{aligned} \quad (1)$$

其中  $V_{\text{applied}}$  是用于抵消电机电阻  $R$  所施加的电压， $L$  为绕组电感， $i$  为电机电流，

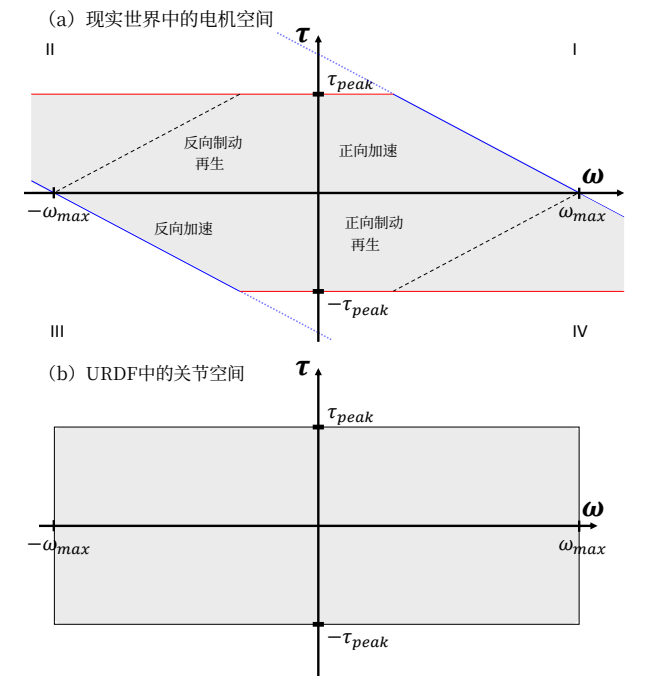


图3. (a) 基于忽略电流时间导数的假设，以灰色显示的BLDC电机工作区域。(b) 由URDF在关节空间中定义的转矩-速度约束。

$V_{\text{BackEMF}}$  是电机旋转时产生的电动势， $K_v$  为角速度常数， $K_t$  为转矩常数。

方程 (1) 描述了在施加电压下电机的转矩与角速度之间的关系。因此，电机的工作区域可以表示为：

$$-V_{\text{bus}} \leq \frac{R}{K_t} \tau + \frac{1}{K_v} \omega \leq V_{\text{bus}}, \quad (2)$$

其中  $V_{\text{bus}}$  为母线电压。该关系由图3 (a) 中蓝色线条之间的区域表示。

为简化起见，一些研究者假设施加的电流与电机转矩呈线性比例关系 [14], [15]。然而，对于因效率和稳定性而被广泛采用的带铁磁芯电机，铁磁芯的磁通饱和会导致非线性特性，如图 2 [12]所示。转矩被限制在峰值扭矩  $\tau_{\text{peak}}$  值以内，该值定义为与线性偏差达到20%时的数值 [16]。由  $\tau_{\text{peak}}$  所限制的电机工作区域如图3(a)中红线之间的区域所示。需要注意的是， $\tau_{\text{peak}}$  并非绝对值，而是一个建议值，不应超过该值以防止绕组因高温而烧毁。当将该电机特性绘制到四个象限中时，第二和第四象限是电机减速的区域，此时电机产生的功的符号为负。从物理意义上解释，电机充当发电机，反电动势可以贡献到总电压中。在这些象限（再生制动或发电机模式）中，工作范围相比第一和第三象限可以更宽，因为电机通过反电动势可能产生超过电池单独所能提供的电压。由此得到的电机工作范围如图3(a)所示。

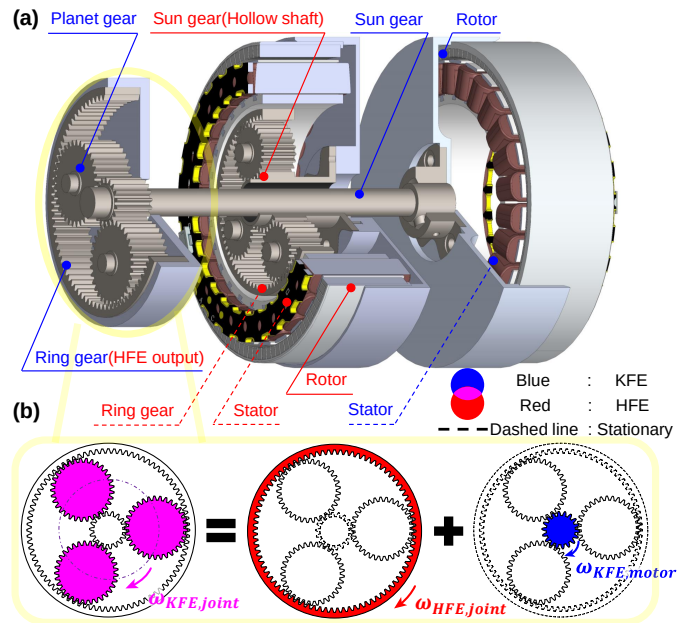


Fig. 4. (a) The conceptual design of KAIST Hound actuator. (b) Superposition of the angular velocity of the KFE and HFE.

In [8], where the  $\tau - \omega$  curves for the motor modules were obtained through experiment, it enabled the precise calculation of actuator constraints for each time step in the simulation for model predictive control (MPC). However, this study did not consider regions besides the first quadrant of the torque-speed graph, which includes deceleration important for achieving dynamic motions.

#### B. Transformation between joint space and motor space

URDF specifies the upper and lower bound of torque and velocity limits of each joint as constants, respectively. These geometries are rectangular, as shown in Fig. 3 (b), and are different from Fig. 3 (a). To reflect the system dynamics from the motor perspective, the parameters of the joint space should be converted to the motor space. URDF uses a joint angle configuration where all parts of the child link move with respect to the ground as much as the parent link moves with respect to the ground. This convention poses difficulty when it describes parallel configurations such as KAIST Hound [9]. As shown in Fig. 4, the hip flexion/extension(HFE) and knee flexion/extension(KFE) modules of KAIST Hound are composed of one piece. Since HFE is designed with a hollow shaft, when HFE rotates, there is no relative motion between the ground and input of the KFE.

The superposition principle was used to obtain the relationship between the input and output of the reducer as a serially connected joint angle configuration from a parallel actuator configuration. In the gearbox of the KFE, the sun gear is the input, the ring gear is stationary, and the planet gear is the output. However, the output part of the HFE is bonded to the ring gear of the KFE. Therefore, each rotation of the HFE input causes the ring gear of the KFE to rotate by  $1/G_{KFE}$ , where  $G_{KFE}$  is the reduction ratio of the KFE joint. The rotation of the KFE ring gear by  $1/G_{KFE}$ , which is the stationary part of the KFE gearbox, denotes that the sun gear rotates by  $-1/G_{KFE}$  from the perspective of the sun

gear. Note that a negative sign indicates the reverse direction. The resulting motor velocity and motor torque obtained by the superposition principle are as follows:

$$\begin{bmatrix} \omega_{HFE,motor} \\ \omega_{KFE,motor} \end{bmatrix} = \begin{bmatrix} \frac{1}{G_{HFE}} & 0 \\ -\frac{1}{G_{HFE}G_{KFE}} & \frac{1}{G_{KFE}} \end{bmatrix}^{-1} \begin{bmatrix} \omega_{HFE,joint} \\ \omega_{KFE,joint} \end{bmatrix} \quad (3)$$

$$\begin{bmatrix} \tau_{HFE,motor} \\ \tau_{KFE,motor} \end{bmatrix} = \begin{bmatrix} \frac{1}{G_{HFE}} & 0 \\ -\frac{1}{G_{HFE}G_{KFE}} & \frac{1}{G_{KFE}} \end{bmatrix}^{-T} \begin{bmatrix} \tau_{HFE,joint} \\ \tau_{KFE,joint} \end{bmatrix} \quad (4)$$

#### C. Motor torque saturation

In the training process, the desired joint position is output from the network and converted to joint torque by the PD controller with the desired joint velocity of zero. Due to the aforementioned limitation that the operating range of the real world motor is represented by a square shape in the joint space of the simulation, the converted joint torque might be infeasible for a real-world motor. Since the joint torque used in the simulation might not be feasible in the real-world motor, the behavior of the robot between the simulation and the real world can be changed. Therefore, we convert the joint torque command that violates the MOR into a motor torque and then clip the value according to equation (1) in the simulation. The saturated motor torque is converted to a saturated joint torque and applied to the robot. One thing to note is that low-level commands correspond to currents in the real world and torques in simulation, respectively. To narrow this gap, the relationship between current and torque was considered. As explained in section. II-A, motors with ferromagnetic cores on the real robot have a non-linear relationship between current and torque. Especially at high torque values, the deviation from linearity becomes larger, so we collected data sets of current and torque from single-axis motors and compensated for the current from quadratic approximation as plotted in Fig. 2.

#### D. Gait reward

The concurrent training of networks and the overall reward functions are based on [17]. However, upon implementing this approach, we observed a phenomenon where motor power was biased towards one side, resulting in the robot's performance being limited by the saturation of one motor. To distribute the torque of the four legs evenly through a symmetrical gait, the gait reward is designed as shown in the TABLE I. The index  $f$  indicates the foot,  $C_{f,i}$  represents the contact state of the foot for index  $i$ . When  $C_{f,i}$  equals 1, the leg is in contact; when  $C_{f,i}$  is 0, the leg is not in contact. When  $i$  is 1, 2, 3, and 4, it refers to the rear right (RR), rear left (RL), front right (FR), and front left (FL) legs, respectively. The effectiveness of the gait reward is demonstrated in the ablation study presented in the TABLE II.

#### E. Learning framework descriptions

1) *Overall structure*: Our framework is described in Fig. 5. The policy network and value network with layers of [256, 128, 64], and the estimator network consists of [256, 128] layers. As in [17], the observation, which is raw data from the

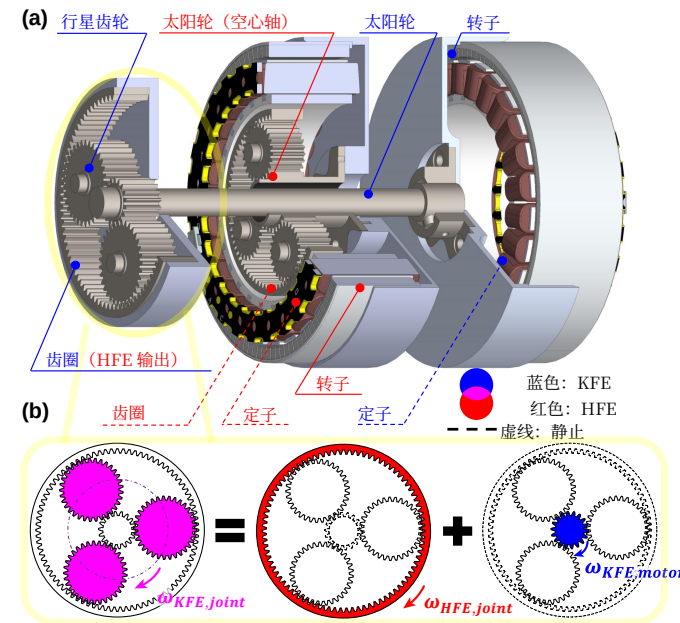


图 4. (a) KAIST Hound 执行器的概念设计。 (b) KFE 和 HFE 角速度的叠加。

在 [8]. 中, 电机模块的  $\tau - \omega$  曲线通过实验获得, 这使得在模型预测控制 (MPC) 的仿真中能够精确计算每个时间步的执行器约束。然而, 该研究未考虑转矩-转速图中第一象限以外的区域, 而这些区域包含了对实现动态运动至关重要的减速过程。

#### B. 关节空间与电机空间之间的变换

URDF 将每个关节的转矩和速度限制的上限和下限分别指定为常数。这些几何形状是矩形的, 如图 3 (b) 所示, 与图 3 (a) 不同。为了从电机角度反映系统动力学, 应将关节空间的参数转换为电机空间。URDF 采用关节角配置, 其中子连杆的所有部分相对于地面的运动量与父连杆相对于地面的运动量相同。这种约定在描述并联构型 (如 KAIST Hound[9]) 时会带来困难。如图 4 所示, KAIST Hound 的髋关节屈曲/伸展 (HFE) 和膝关节屈曲/伸展 (KFE) 模块由一体件组成。由于 HFE 采用空心轴设计, 当 HFE 旋转时, 地面与 KFE 的输入之间没有相对运动。

利用叠加原理, 从并联执行器配置中推导出减速器输入与输出之间的关系, 即串联关节角配置。在 KFE 的齿轮箱中, 太阳轮为输入, 齿圈静止, 行星齿轮为输出。然而, HFE 的输出部分与 KFE 的齿圈相连。因此, HFE 输入的每次旋转都会使 KFE 的齿圈旋转  $1/G_{KFE}$ , 其中  $G_{KFE}$  是 KFE 关节的减速比。KFE 齿圈 (即 KFE 齿轮箱的静止部分) 旋转  $1/G_{KFE}$ , 表示从太阳轮的角度来看, 太阳轮旋转了  $-1/G_{KFE}$ 。

需要注意的是, 负号表示反向。通过叠加原理得到的电机速度和电机转矩如下:

$$\begin{bmatrix} \omega_{HFE,motor} \\ \omega_{KFE,motor} \end{bmatrix} = \begin{bmatrix} \frac{1}{G_{HFE}} & 0 \\ -\frac{1}{G_{HFE}G_{KFE}} & \frac{1}{G_{KFE}} \end{bmatrix}^{-1} \begin{bmatrix} \omega_{HFE,joint} \\ \omega_{KFE,joint} \end{bmatrix} \quad (3)$$

$$\begin{bmatrix} \tau_{HFE,motor} \\ \tau_{KFE,motor} \end{bmatrix} = \begin{bmatrix} \frac{1}{G_{HFE}} & 0 \\ -\frac{1}{G_{HFE}G_{KFE}} & \frac{1}{G_{KFE}} \end{bmatrix}^{-T} \begin{bmatrix} \tau_{HFE,joint} \\ \tau_{KFE,joint} \end{bmatrix} \quad (4)$$

#### C. 电机转矩饱和

在训练过程中, 期望关节位置由网络输出, 并通过 PD 控制器在期望关节速度为零的情况下转换为关节转矩。由于前述限制, 真实世界电机的工作范围在仿真关节空间中呈方形, 转换后的关节转矩可能对真实世界电机不可行。由于仿真中使用的关节转矩在真实世界电机中可能不可行, 机器人在仿真与真实世界之间的行为可能会发生变化。因此, 我们在仿真中将违反 MOR 的关节转矩指令转换为电机转矩, 然后根据公式 (1) 对数值进行裁剪。饱和后的电机转矩被转换为饱和关节转矩并施加到机器人上。需要注意的是, 低级指令分别对应真实世界中的电流和仿真中的转矩。为缩小这一差距, 我们考虑了电流与转矩之间的关系。如第 II-A 节所述, 真实机器人上带有铁磁芯的电机, 其电流与转矩之间呈非线性关系。尤其是在高转矩值下, 与线性关系的偏差会变大, 因此我们从单轴电机上收集了电流和转矩数据集, 并通过二次近似对电流进行了补偿, 如图 2 所示。

#### D. 步态奖励

网络和整体奖励函数的并行训练基于 [17]。然而, 在实施该方法后, 我们观察到电机功率偏向一侧的现象, 导致机器人性能受到单个电机饱和的限制。为了通过对称步态均匀分配四条腿的转矩, 步态奖励的设计如表 I 所示。索引  $f$  表示足,  $C_{f,i}$  表示索引  $i$  的足的接触状态。当  $C_{f,i}$  等于 1 时, 腿处于接触状态; 当  $C_{f,i}$  为 0 时, 腿未接触。当  $i$  为 1、2、3 和 4 时, 分别指右后 (RR)、左后 (RL)、右前 (FR) 和左前 (FL) 腿。步态奖励的有效性在表 II 所示的消融研究中得到验证。

#### E. 学习框架描述

1) 总体结构: 我们的框架如图 5 所示。策略网络和价值网络具有 [256, 128, 64] 层, 估计器网络由 [256, 128] 层组成。如 [17], 所述, 观测数据 (即来自传感器的原始数据) 输入估计器网络。来自估计器的机体线速度、足部高度和接触概率与原始数据合并后输入演员网络。演员网络输出的期望关节位置输入关节 PD 控制器, 其参数为  $K$  [256, N 128, 64],  $m$  [256, 128] rad [17], 和  $KN$  m s rad, 并转换为关节转矩。



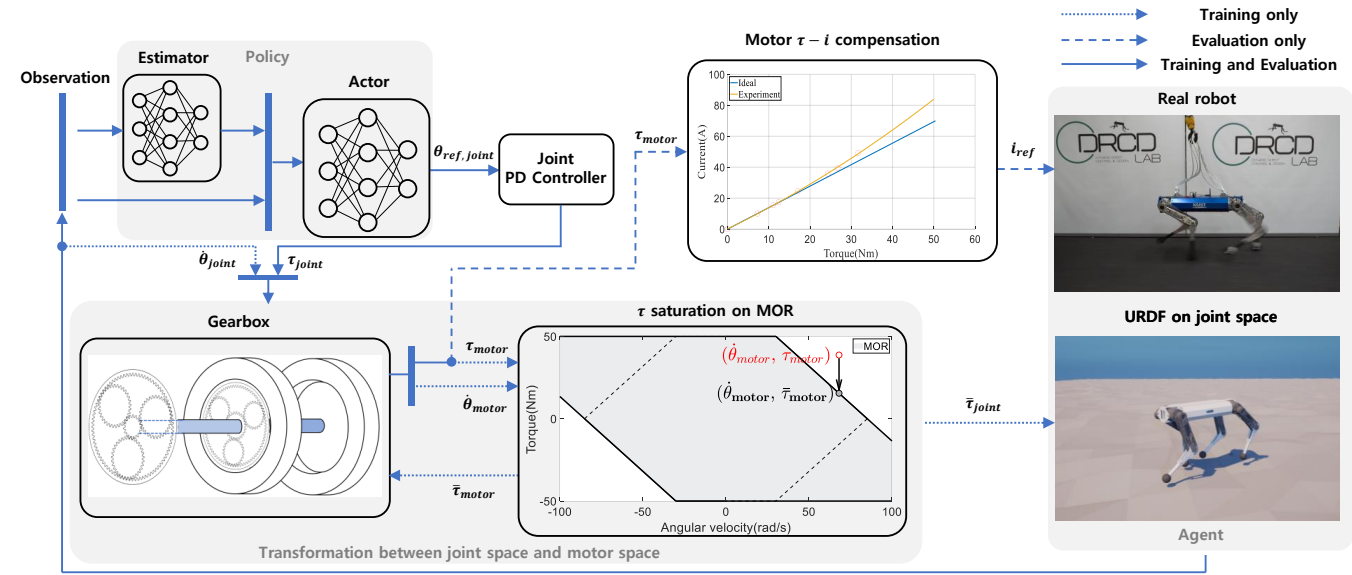


Fig. 5. The overall proposed reinforcement learning framework is shown. The action was expressed in terms of motor torque, and the torque was saturated by the MOR in the simulation. The nonlinearity of the current and torque due to the ferromagnetic core is compensated when the network performs inference.

sensor, inputs the estimator network. The body linear velocity, foot height, and contact probability from the estimator are merged with the raw data and input to the actor. The desired joint position from the actor is input to the joint PD controller with  $K_p = 50 \text{ N} \cdot \text{m} \cdot \text{rad}^{-1}$  and  $K_d = 1.0 \text{ N} \cdot \text{m} \cdot \text{s} \cdot \text{rad}^{-1}$  and converted to joint torque. The joint velocity and joint torque commands are converted through equations (3)-(4) and saturated to the boundary of MOR if the motor torque corresponding to the motor velocity violates the MOR. The saturated motor torque is converted to joint space by the gearbox and input to the robot. In the real world, the motor torque obtained from the network is converted to the input current of the motor by corresponding to the  $\tau-i$  graph as shown in Fig. 5.

2) *Reward functions*: Our reward functions are defined as shown in the TABLE I. Bolded text indicates a new term added for high-speed running. Note that  $k_{\text{contact}}$  is only 1 in the time step immediately before the contact occurs and 0 otherwise.

We added a contact velocity reward that reduces the relative velocity between the ground and the foot just before the collision. It is designed to prevent unstable contact caused by the restitution coefficient between the ground and the foot, which is not observed in the simulation based on the hard contact model, RaiSim [18].

3) *Strategies for high-speed running in simulation*: The body mass, inertia, and center of mass positions, which are difficult to identify, were randomized. We also implemented a curriculum-based approach, progressively increasing the velocity command in the forward direction, which was uniformly sampled from  $U^1(-0.3, 1.5) \text{ m/s}$  at the beginning. At the end of the learning process, this range was broadened to  $U^1(-1.4, 7.0) \text{ m/s}$  as shown in Fig. 12. At first, to allow the robot to react to abrupt decelerations and accelerations, we saved the robots final state from the previous episode and used it as the initial state for the next episode with a 25% probability. We observed that when the gap between the robots

TABLE I  
REWARD FUNCTIONS

| Reward                  | Expression                                                                                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Linear velocity         | $C_V \exp(-  V_{xy}^{\text{desired}} - V_{xy}  ^2)$                                                                                                                                                                                            |
| Angular velocity        | $C_\theta \exp(-1.5(\dot{\theta}_z^{\text{desired}} - \dot{\theta}_z)^2)$                                                                                                                                                                      |
| <b>Contact velocity</b> | $C_{\text{contact}} k_{\text{contact}}   V_{f,z,i}  ^2$                                                                                                                                                                                        |
| Foot slip               | $C_{\text{slip}} C_{f,i}   V_{f,xy,i}  ^2$                                                                                                                                                                                                     |
| Foot clearance          | $C_{\text{clearance}} (-C_{f,i})(w_{p,f,z,i} - w_{p,f,z}^{\text{desired}})^2   V_{f,xy,i}  ^{0.5}$                                                                                                                                             |
| Orientation             | $C_{\text{orientation}} (\text{angle}(\phi_{\text{body},z}, \phi_{\text{world},z}))^2$                                                                                                                                                         |
| Joint torque            | $C_\tau   \tau  ^2$                                                                                                                                                                                                                            |
| Joint position          | $C_q   q_t - q_{\text{nominal}}  ^2$                                                                                                                                                                                                           |
| Joint speed             | $C_{\dot{q}}   \dot{q}_t  ^2$                                                                                                                                                                                                                  |
| Joint acceleration      | $C_{\ddot{q}}   \ddot{q}_t - \ddot{q}_{t-1}  ^2$                                                                                                                                                                                               |
| Action smoothness 1     | $C_{\text{smoothness1}}   q_t^{\text{desired}} - q_{t-1}^{\text{desired}}  ^2$                                                                                                                                                                 |
| Action smoothness 2     | $C_{\text{smoothness2}}   q_t^{\text{desired}} - 2q_{t-1}^{\text{desired}} + q_{t-2}^{\text{desired}}  ^2$                                                                                                                                     |
| Base motion             | $C_{\text{base}} (0.8V_z^2 + 0.2 \dot{\theta}_x  + 0.2 \dot{\theta}_y )$                                                                                                                                                                       |
| <b>Gait</b>             | $\begin{cases} \text{if } C_{f,1} \wedge C_{f,4} \wedge \neg C_{f,2} \wedge \neg C_{f,3} : C_{\text{gait}} \\ \text{else if } \neg C_{f,1} \wedge \neg C_{f,4} \wedge C_{f,2} \wedge C_{f,3} : C_{\text{gait}} \\ \text{else} : 0 \end{cases}$ |

| Reward Coefficients      |          |                    |         |                          |       |
|--------------------------|----------|--------------------|---------|--------------------------|-------|
| $C_V$                    | 6.0      | $C_{\dot{\theta}}$ | 3.0     | $C_{\text{gait}}$        | 1.2   |
| $C_{\text{contact}}$     | -6.0     | $C_{\text{slip}}$  | -1.6e-1 | $C_{\text{orientation}}$ | -3.0  |
| $C_\tau$                 | -2e-4    | $C_q$              | -0.75   | $C_{\dot{q}}$            | -3e-4 |
| $C_{\ddot{q}}$           | -0.67e-2 | $C_{\text{base}}$  | -4      | $C_{\text{smoothness1}}$ | -2.5  |
| $C_{\text{smoothness2}}$ | -0.8     |                    |         |                          |       |

current velocity and the command velocity was too large, the robot would go to a standstill to avoid terminating the episode while tracking. Therefore, when the episode starts from a 75% stop, the command is increased gradually over the first second of the episode. For the remaining 25%, we added an additional condition to ensure that the command velocity difference between the previous episode and the current episode does not exceed 2 m/s. As the robot runs at higher speeds, it receives larger penalties due to a reward term with a negative coefficient, such as joint torque and joint speed. Therefore, as the robot reaches higher speeds, it increasingly tends to abandon command tracking. To encourage the robot to follow high-speed commands, the weights of linear velocity and angular velocity in the reward term were gradually increased.

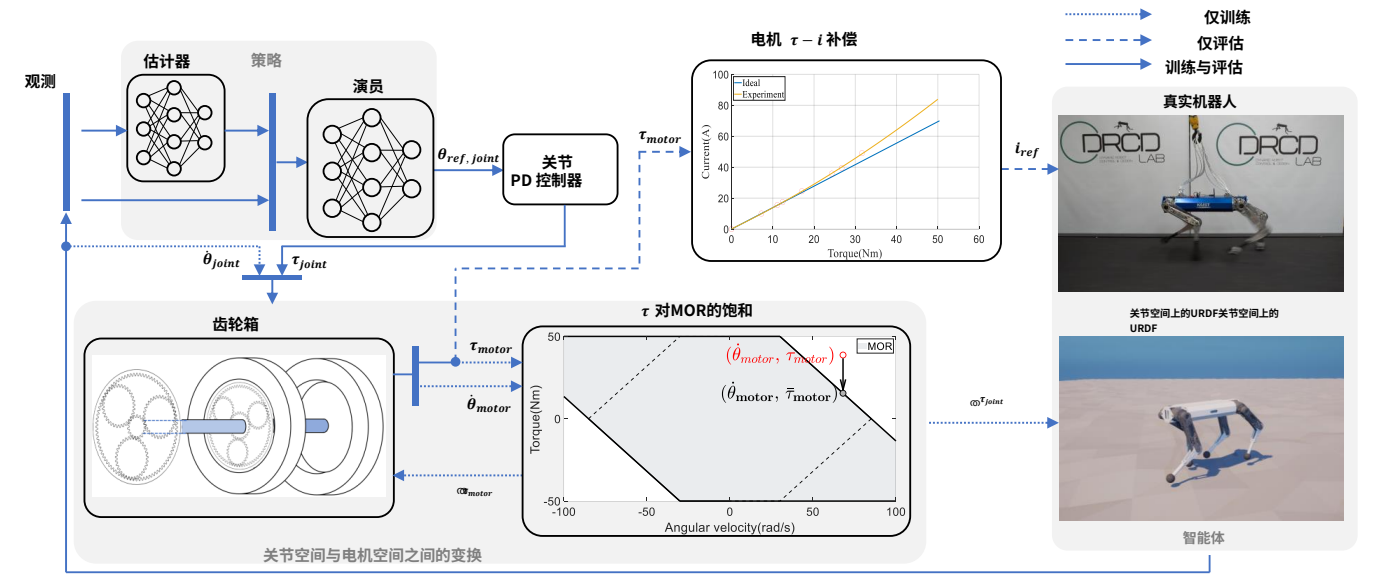


图5. 展示了整体提出的强化学习框架。动作以电机转矩的形式表示，并在仿真中通过MOR对转矩进行饱和和处理。由于铁磁芯导致的电流与转矩非线性，在网络进行推理时得到了补偿。

关节速度和关节转矩指令通过公式(3)-(4)进行转换，如果对应电机速度的电机转矩违反MOR，则将其饱和至MOR边界。饱和后的电机转矩通过齿轮箱转换到关节空间并输入机器人。在真实世界中，从网络获得的电机转矩通过对应的  $\tau-i$  曲线（如图5所示）转换为电机的输入电流。

2) 奖励函数：我们的奖励函数如表I所示。加粗文本表示针对高速奔跑新增的项。请注意， $k_{\text{contact}}$  仅在接触发生前的时间步为1，其余时间均为0。

我们新增了接触速度奖励，用于减小碰撞前地面与足之间的相对速度。该奖励旨在防止因地面与足之间的恢复系数导致的不稳定接触，而这种接触在基于硬接触模型的仿真RaiSim [18]中无法观察到。

3) 仿真中高速奔跑的策略：对难以辨识的机体质量、惯性及质心位置进行了随机化处理。我们还实施了基于课程的方法，逐步提高前进方向的速度指令，该指令在初始阶段从 $U^1(-0.3, 1.5) \text{ m/s}$ 均匀采样。在学习过程结束时，该范围扩展至 $U^1(-1.4, 7.0) \text{ m/s}$ ，如图12所示。起初，为了让机器人能够应对突然的减速和加速，我们以25%的概率保存机器人上一回合的最终状态，并将其作为下一回合的初始状态。我们观察到，当机器人

表I 奖励函数

| 奖励                       | 表达式                                                                                                                                                                                                                                               |                    |         |                          |       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------|--------------------------|-------|
| 线速度                      | $C_V \text{实验}(-  V_{xy}^{\text{desired}} - V_{xy}  ^2)$                                                                                                                                                                                          |                    |         |                          |       |
| 角速度                      | $C_\theta \exp(-1.5(\dot{\theta}_z^{\text{desired}} - \dot{\theta}_z)^2)$                                                                                                                                                                         |                    |         |                          |       |
| 接触速度                     | $C_{\text{contact}} k_{\text{contact}}   V_{f,z,i}  ^2$                                                                                                                                                                                           |                    |         |                          |       |
| 足部滑动                     | $C_{\text{slip}} C_{f,i}   V_{f,xy,i}  ^2$                                                                                                                                                                                                        |                    |         |                          |       |
| 足部离地高度                   | $C_{\text{clearance}} (-C_{f,i})(w_{p,f,z,i} - w_{p,f,z}^{\text{desired}})^2   V_{f,xy,i}  ^{0.5}$                                                                                                                                                |                    |         |                          |       |
| 朝向                       | $C_{\text{orientation}} (\text{角度}(\varphi_{\text{body},z}, \varphi_{\text{world},z}))^2$                                                                                                                                                         |                    |         |                          |       |
| 关节转矩                     | $C_\tau   \tau  ^2$                                                                                                                                                                                                                               |                    |         |                          |       |
| 关节位置                     | $C_q   q_t - q_{\text{nominal}}  ^2$                                                                                                                                                                                                              |                    |         |                          |       |
| 关节速度                     | $C_{\dot{q}}   \dot{q}_t  ^2$                                                                                                                                                                                                                     |                    |         |                          |       |
| 关节加速度                    | $C_{\ddot{q}}   \ddot{q}_t - \ddot{q}_{t-1}  ^2$                                                                                                                                                                                                  |                    |         |                          |       |
| 动作平滑度1                   | $C_{\text{smoothness1}}   q_t^{\text{desired}} - q_{t-1}^{\text{desired}}  ^2$                                                                                                                                                                    |                    |         |                          |       |
| 动作平滑度2                   | $C_{\text{smoothness2}}   q_t^{\text{desired}} - 2q_{t-1}^{\text{desired}} + q_{t-2}^{\text{desired}}  ^2$                                                                                                                                        |                    |         |                          |       |
| 底座运动                     | $C_{\text{base}} (0.8V_z^2 + 0.2 \dot{\theta}_x  + 0.2 \dot{\theta}_y )$                                                                                                                                                                          |                    |         |                          |       |
| Gait                     | $\begin{cases} \text{如果 } C_{f,1} \wedge C_{f,4} \wedge \neg C_{f,2} \wedge \neg C_{f,3} : C_{\text{gait}} \text{否} \\ \text{则如果 } \neg C_{f,1} \wedge \neg C_{f,4} \wedge C_{f,2} \wedge C_{f,3} : C_{\text{gait}} \\ \text{否则: } 0 \end{cases}$ |                    |         |                          |       |
|                          |                                                                                                                                                                                                                                                   |                    |         |                          |       |
| 奖励系数                     |                                                                                                                                                                                                                                                   |                    |         |                          |       |
| $C_V$                    | 6.0                                                                                                                                                                                                                                               | $C_{\dot{\theta}}$ | 3.0     | $C_{\text{gait}}$        | 1.2   |
| $C_{\text{contact}}$     | -6.0                                                                                                                                                                                                                                              | $C_{\text{slip}}$  | -1.6e-1 | $C_{\text{orientation}}$ | -3.0  |
| $C_\tau$                 | -2e-4                                                                                                                                                                                                                                             | $C_q$              | -0.75   | $C_{\dot{q}}$            | -3e-4 |
| $C_{\ddot{q}}$           | -0.67e-2                                                                                                                                                                                                                                          | $C_{\text{base}}$  | -4      | $C_{\text{smoothness1}}$ | -2.5  |
| $C_{\text{smoothness2}}$ | -0.8                                                                                                                                                                                                                                              |                    |         |                          |       |

当前速度与指令速度之间的差距过大时，机器人会进入停止状态，以避免在跟踪过程中提前终止回合。因此，当回合从75%的停止状态开始时，指令会在回合的第一秒内逐渐增加。对于剩余的25%，我们添加了一个额外条件，以确保前一回合并当前回合之间的指令速度差不超过2 m/s。随着机器人以更高的速度奔跑，它会因带有负系数的奖励项（如关节转矩和关节速度）而受到更大的惩罚。因此，随着机器人达到更高的速度，它越来越倾向于放弃指令跟踪。为了鼓励机器人遵循高速指令，奖励项中线速度和角速度的权重被逐渐增加。



Fig. 6. Customized foot for high-speed running. Left : The original design in a cylindrical shape. Right : A lightweight design in a part of a spherical shape.

#### F. Lightweight foot design

We designed a specialized foot for high-speed running with reduced leg inertia, which can decrease impact forces and increase mechanical bandwidth [19], [20]. Since the foot is located at the end of the leg link, a lightweight design is important to reduce the moment of inertia of the knee joint. Most quadruped robots are designed with feet that are either spherical or cylindrical in shape [9], [21]–[27], which allow for a constant rolling contact that helps simplify contact dynamics to a point or line. However, as the foot approaches a perfect sphere or cylinder shape, the weight also increases. In the case of high-speed running, the robot only needs a foot shape that corresponds to the attack angle and detach angle between the flat ground and the foot. Therefore, we investigated the running data in simulation and designed the foot with a minimal sphere shape that ensures contact with the ground. As a result, the mass of the calf was reduced to 38.0%, and the inertia in the pitch direction was reduced to 37.4% of its original.

### III. RESULTS

#### A. Evaluation in simulation

We investigated the effectiveness of MOR constraints for robot performance in simulation. First, we trained a controller without MOR constraints as a baseline. The robot reached the target speed from rest with an acceleration of  $1 \text{ m/s}^2$ . The motor operating region is violated when running at high speed in the simulation, as shown by the red line in Fig. 8 (c). To reflect the behavior of actual motors in the simulation, we clipped the torque commands that violate the MOR, within the operational limits.

We used average reward as our performance metric, which indicates how close the robot’s locomotion is to our designed behavior. The average reward was computed for 3 seconds after the robot speed converged. The result is shown in Fig. 7. Note that  $\Delta \text{reward}_{MOR} = (\text{Policy evaluation without MOR}) - (\text{Policy evaluation with MOR})$ . As the speed increases, the average reward, including command tracking performance, drops under the MOR constraint. Since the velocity below 3 m/s was achieved with torque commands inside the MOR, there was no performance drop.

In the real world, contact probabilities from a state estimation network have been observed to be noisy at high speeds. Therefore, rewards that require contact information are difficult to use as a metric for robot performance. To

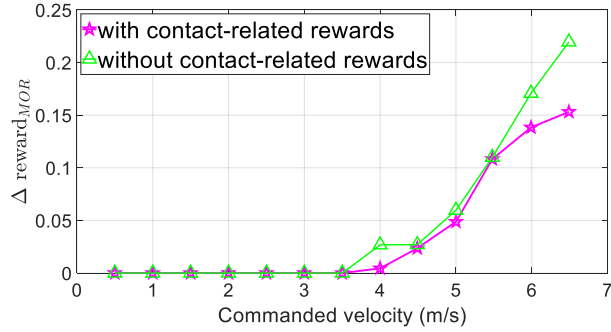


Fig. 7. The graph shows that when the policy learned without a MOR constraint is evaluated with a MOR constraint in the simulation, the performance drop occurs from 4 m/s using torque that violates the MOR. The two lines exhibit a similar trend in  $\Delta \text{reward}_{MOR}$ , even when excluding contact-related rewards that are challenging to estimate in the real world.

address the aforementioned problem, we also investigated the difference of average reward without contact-related rewards. It showed a trend where the gap becomes larger as the speed increases, similar to the difference in average reward. These results show when the trained network without MOR constraint is evaluated in the real world, the torque output limitation can cause a performance drop.

#### B. System setup

We used an AMD Ryzen Threadripper PRO 5995WX and a NVIDIA GeForce RTX 3080 Ti for the training process. The control policy requires about 6 hours to complete its training and operates at a frequency of 100 Hz. The on-board PC for control policy and estimation algorithms operates on a ThinkStation P350 Tiny. State estimation for body orientation is conducted by [28]. User inputs are transmitted to the high-level PC in real time via a Taranis X9D Plus transmitter with a X8R receiver. A commercial motor controller, Elmo Platinum Twitter 100 V/70 A, communicates with the high-level PC at a rate of 2 kHz through EtherCAT. The robot is powered by two different types of Lithium-Ion battery packs. For driving the motor, a 21700 Li-ion battery pack configured in a 20S 2P, 20 batteries in series, and two sets of these in parallel, is employed, offering high power within the limited body space. The other electrical parts are powered by a separate 18650 Li-ion battery pack in a 6S 3P configuration. The motor battery was charged to 81 V with a margin from the maximum charge of 84 V to permit reverse current from the motor.

#### C. Experimental results

TABLE II  
ABLATION STUDY FOR THE REAL-ROBOT

|                      | without<br>gait reward | without<br>customized foot | without<br>MOR constraint | <b>Ours</b> |
|----------------------|------------------------|----------------------------|---------------------------|-------------|
| Best record<br>(m/s) | 6.0                    | 5.5                        | 4.5                       | <b>6.5</b>  |

1) *Indoor experiments*: We investigated how three major factors affect the performance of high-speed running.

- Without gait reward: To verify the performance drop caused by the saturation of the power of single-motor, we set the coefficient of gait reward to zero. The gait



图6. 用于高速奔跑的定制足部。左：圆柱形的原始设计。右：部分球形的轻量化设计。

#### F. 轻量化足部设计

我们设计了一种专门用于高速奔跑的足部，以降低腿部惯性，这可以减少冲击力并提高机械带宽 [19], [20]。由于足部位于腿杆末端，轻量化设计对于减少膝关节的转动惯量至关重要。大多数四足机器人的足部设计为球形或圆柱形 [9], [21]–[27]，这样可以实现恒定的滚动接触，有助于将接触动力学简化为点或线。然而，随着足部接近完美的球体或圆柱体形状，重量也会增加。在高速奔跑的情况下，机器人只需要一种与平坦地面之间的攻角和分离角相对应的足部形状。因此，我们研究了仿真中的奔跑数据，并设计了具有最小球面形状的足部，以确保与地面接触。最终，小腿的质量减少到原来的38.0%，俯仰方向上的惯性减少到原来的37.4%。

### III. 结果

#### A. 仿真评估

我们在仿真中研究了电机运行区域约束对机器人性能的有效性。首先，我们训练了一个不带电机运行区域约束的控制器作为基线。机器人以  $1 \text{ m/s}^2$  的加速度从静止状态达到目标速度。在仿真中，高速奔跑时电机工作区域被超出，如图8 (c) 中的红线所示。为了在仿真中反映实际电机的行为，我们在运行极限内对违反电机运行区域约束的力矩指令进行了截断。

我们使用平均奖励作为性能指标，它表示机器人的运动与我们设计行为的接近程度。平均奖励在机器人速度收敛后计算3秒。结果如图7所示。注意， $\Delta \text{奖励}_{MOR} = (\text{无电机运行区域约束的策略评估}) - (\text{有电机运行区域约束的策略评估})$ 。随着速度的增加，包括指令跟踪性能在内的平均奖励在电机运行区域约束下有所下降。由于低于3 m/s的速度是在电机运行区域内的力矩指令下实现的，因此没有出现性能下降。

在真实世界中，来自状态估计网络的接触概率在高速情况下已被观察到存在噪声。因此，需要接触信息的奖励难以作为机器人性能的度量指标。为了

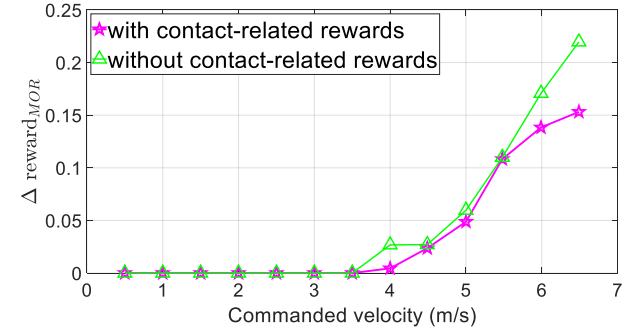


图7. 该图显示，当在仿真中未使用MOR约束训练的策略在MOR约束下进行评估时，从4 m/s开始出现性能下降，此时使用了违反MOR约束的转矩。两条曲线在  $\Delta \text{奖励}_{MOR}$  上表现出相似的趋势，即使排除了在真实世界中难以估计的接触相关奖励也是如此。

解决上述问题，我们还研究了无接触相关奖励时平均奖励的差异。结果显示，随着速度增加，差距变大的趋势与平均奖励的差异相似。这些结果表明，当未经MOR约束训练的网络在真实世界中评估时，力矩输出限制可能导致性能下降。

#### B. 系统设置

我们使用AMD Ryzen Threadripper PRO 5995WX和NVIDIA GeForce RTX 3080 Ti进行训练过程。控制策略大约需要6小时完成训练，并以100 Hz的频率运行。用于控制策略和估计算法的机载PC运行在ThinkStation P350 Tiny上。身体姿态的状态估计由 [28] 完成。用户输入通过Taranis X9D Plus发射器和X8R接收机实时传输到高层PC。商用电机控制器Elmo Platinum Twitter 100 V/70 A通过EtherCAT以2 kHz的速率与高层PC通信。机器人由两种不同类型的锂离子电池组供电。驱动电机采用21700锂离子电池组，配置为20S 2P，即20节电池串联，两组并联，在有限的车身空间内提供高功率。其他电子部件由独立的18650锂离子电池组供电，配置为6S 3P。电机电池充电至81 V，与最大充电电压84 V保持一定余量，以允许电机产生反向电流。

#### C. 实验结果

表II 真实机器人消融研究

|                   | 无<br>步态奖励 | 无<br>定制足部 | 无<br>MOR约束 | <b>Ours</b> |
|-------------------|-----------|-----------|------------|-------------|
| 最佳记录<br>(米/<br>秒) | 6.0       | 5.5       | 4.5        | <b>6.5</b>  |

1) 室内实验：我们研究了三个主要因素如何影响高速奔跑的表现。

- 无步态奖励：为验证单电机功率饱和导致的性能下降，我们将步态奖励的系数设为零。步态



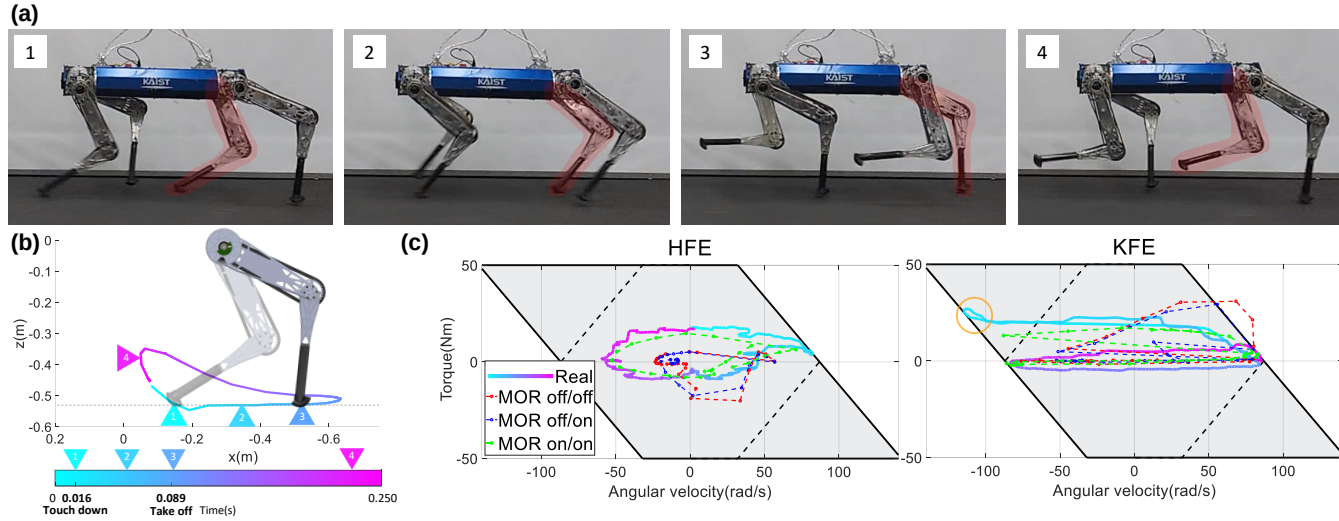


Fig. 8. These figures provide detailed data of the rear right leg collected while running at 6.5 m/s in one period. (a): The red shading in the four snapshots represents the movement of the rear right leg. (b): Foot trajectory relative to the shoulder joint. The one cycle is 0.25 seconds, and the color gradually changes from cyan to magenta to indicate the position of the trajectory over time. (c): The  $\tau - \omega$  trajectories for 6.5m/s running in simulation and real-world. MOR off/off: Training without MOR constraints and evaluation without MOR constraints. MOR off/on: Training without MOR constraints and evaluation with MOR constraints. MOR on/on: Training with MOR constraints and evaluation with MOR constraints.

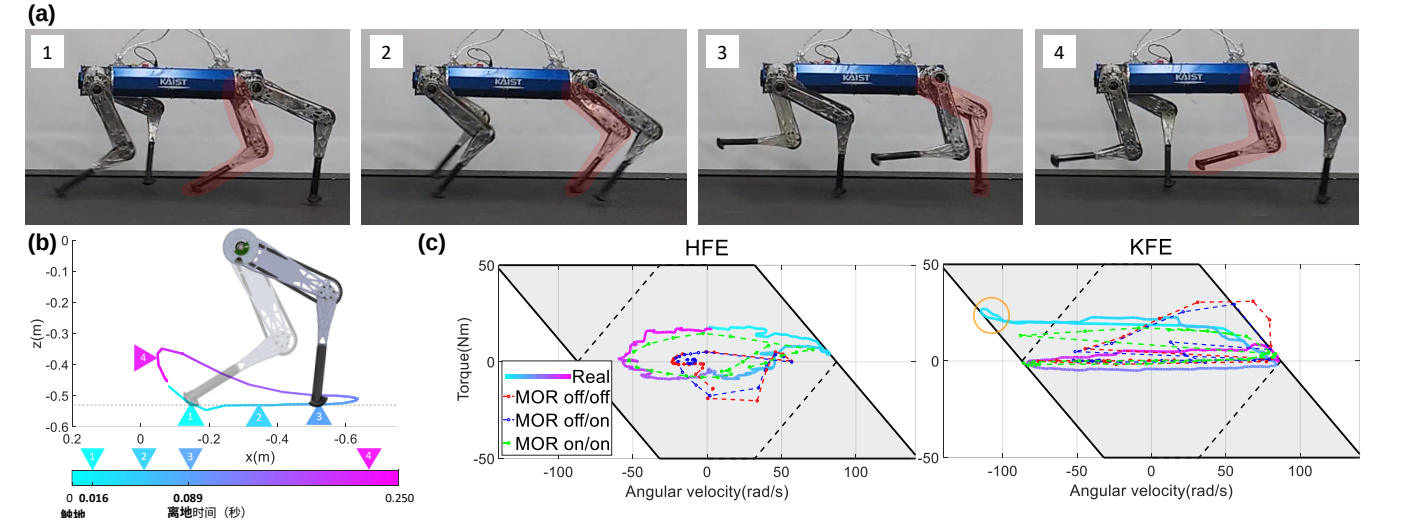


图8. 这些图提供了以6.5米/秒奔跑一个周期内右后腿的详细数据。(a)：四个快照中的红色阴影表示右后腿的运动。(b)：相对于肩关节的足部轨迹。一个周期为0.25秒，颜色从青色逐渐变为品红色，以指示轨迹随时间变化的位置。(c)： $\tau - \omega$  仿真和现实世界中6.5米/秒奔跑的轨迹。MOR关闭/关闭：无MOR约束训练和无MOR约束评估。MOR关闭/开启：无MOR约束训练和有MOR约束评估。MOR开启/开启：有MOR约束训练和有MOR约束评估。

reward is designed to induce an even distribution of the motor's power through a symmetrical gait.

- Without customized foot: To verify whether the reduced inertia due to the lightweight design of the foot is effective for high-speed running, we train the robot model with an original foot, which is cylindrical shape. The learned policy is used to control a robot with an original foot to measure its speed.
- Without MOR constraint: To validate the effect of the MOR constraint, we exclude clipping for torque commands that violate the MOR in the training process.

The robot ran at 0.5 m/s intervals on a treadmill with the trained controllers. We considered the trial at each speed interval a success if the robot maintained that speed on the treadmill for 10 seconds. As shown in table II, the controller trained with our proposed method achieved the highest speed of 6.5 m/s. When we increased the treadmill speed to 7 m/s, the robot was pushed backward, and its feet went off the treadmill, so 7 m/s was considered a failure. The other three controllers recorded the speed of 6.0, 5.5, and 4.5 m/s respectively which are slower than 6.5 m/s.

To further demonstrate the effectiveness of the MOR constraint, we computed the reward gap between the simulation and the real world from two policies. The two policies were trained with the MOR constraint and without the MOR constraint, respectively. Note that both policies can achieve 6.5 m/s in simulation.  $\Delta \text{reward}_{sim-to-real} = (\text{policy evaluation in simulation}) - (\text{policy evaluation in the real world})$  is defined as a performance metric, excluding the term computed by contact information, which is noisy at high speeds. Since the robot runs on a treadmill, it is difficult to measure linear velocity with a motion capture system. Therefore, we use the linear velocity output from the estimator network trained by supervised learning with the true value in simulation and representing low estimation error to calculate the reward [17].

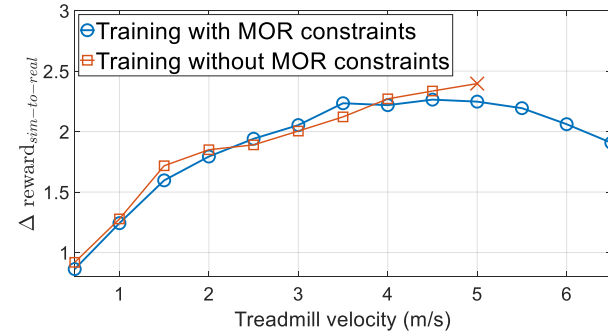


Fig. 9. The graph shows that applying the MOR constraint reduces the average reward when evaluating a policy learned without the MOR constraint in simulation. The controller trained without MOR constraints fell while the robot was running at 5 m/s.

The results are shown in Fig. 9. In the real world, as the robot runs faster, the higher  $\Delta \text{reward}_{sim-to-real}$  due to the increase in unmodeled dynamics [29], [30] that are not included in the simulation. As shown in Fig. 7, the  $\Delta \text{reward}_{MOR}$  of the two policies tends to differ at high speeds above 3.5 m/s, where the motor uses torque that violates the MOR.

For the policy trained with MOR constraints,  $\Delta \text{reward}_{sim-to-real}$  does not increase after 3.5 m/s. However, the robot controlled by the learned policy without MOR constraints continued to increase in  $\Delta \text{reward}_{sim-to-real}$  and fell at 5 m/s. This result shows that enforcement of actuator limits helps the simulation behavior resemble the real world when the motor uses high power.

2) *Analysis at the top speed:* At a running speed of 6.5 m/s, the cost of transport (CoT) for 15 strides, calculated using mechanical power and joule heating, is 0.29. Fig. 8 (a) shows snapshots of the four distinct behaviors during one cycle at top speed. The foot trajectory of the rear right leg over time is plotted in Fig. 8 (b) as a change in color, and the  $\tau - \omega$  trajectories of the motor are plotted in Fig. 8 (c). The dashed line shows the values from the simulation, and

奖励旨在通过非对称步态实现电机功率的均匀分配。

- 无定制足部：为了验证轻量化设计带来的足部惯性减小是否对高速奔跑有效，我们使用原始足部（圆柱形）训练机器人模型。学习策略被用于控制带有原始足部的机器人，以测量其速度。
- 无MOR约束：为验证MOR约束的效果，我们在训练过程中排除了对违反MOR的转矩指令的裁剪。

机器人在跑步机上以0.5 m/s的速度间隔运行，并使用训练控制器进行控制。如果机器人在跑步机上保持该速度10秒，我们则认为该速度区间的试验成功。如表II所示，采用我们所提方法训练的控制器达到了6.5 m/s的最高速度。当我们将跑步机速度提高到7 m/s时，机器人被向后推，其脚部离开了跑步机，因此7 m/s被视为失败。其他三个控制器分别记录了6.0、5.5和4.5 m/s的速度，均低于6.5 m/s。

为了进一步证明MOR约束的有效性，我们计算了两种策略在仿真与真实世界之间的奖励差距。这两种策略分别在有MOR约束和无MOR约束的条件下进行训练。需要注意的是，两种策略在仿真中均能达到6.5 m/s的速度。  $\Delta \text{奖励}_{sim-to-real} = (\text{仿真中的策略评估}) - (\text{真实世界中的策略评估})$  被定义为性能指标，其中不包括由接触信息计算得到的项，因为接触信息在高速时噪声较大。由于机器人在跑步机上运行，使用动作捕捉系统难以测量线速度。因此，我们使用由监督学习训练的估计器网络输出的线速度来计算奖励 [17]，该网络在仿真中使用真值进行训练，且估计误差较低。

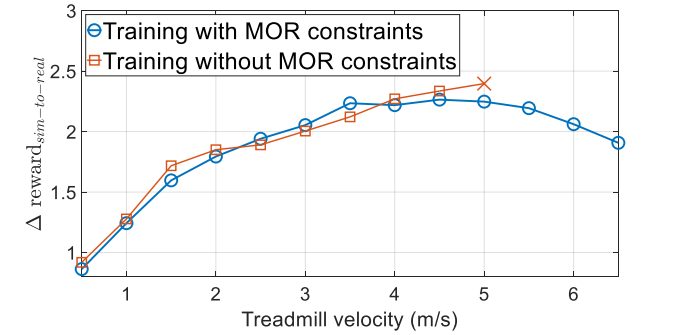


图9. 该图显示，在仿真中评估未施加MOR约束训练的策略时，应用MOR约束会降低平均奖励。未施加MOR约束训练的控制器在机器人以5 m/s速度运行时摔倒。

结果如图9所示。在真实世界中，随着机器人运行速度的加快，由于未建模动力学的增加， $\Delta \text{奖励}_{sim-to-real}$  也随之增大 [29], [30]，而这些未建模动力学并未包含在仿真中。如图7所示，在高于3.5 m/s的高速下，两种策略的  $\Delta \text{奖励}_{MOR}$  趋于不同，此时电机使用的转矩违反了MOR约束。

对于使用MOR约束训练的策略， $\Delta \text{奖励}_{sim-to-real}$  在3.5 m/s后不再增加。然而，由未施加MOR约束的学习策略控制的机器人在  $\Delta \text{奖励}_{sim-to-real}$  方面持续增加，并在5 m/s时摔倒。这一结果表明，当电机使用高功率时，执行器限制的强制执行有助于仿真行为更接近真实世界。

2) 最高速度下的分析：在6.5 m/s的跑步速度下，使用机械功率和焦耳热计算的15个步幅的运输成本 (CoT) 为0.29。图8 (a) 展示了最高速度下一个周期内四种不同行为的快照。右后腿的足部轨迹随时间的变化绘制在图8 (b) 中，以颜色变化表示，电机的  $\tau - \omega$  轨迹绘制在图8 (c) 中。虚线显示仿真值，



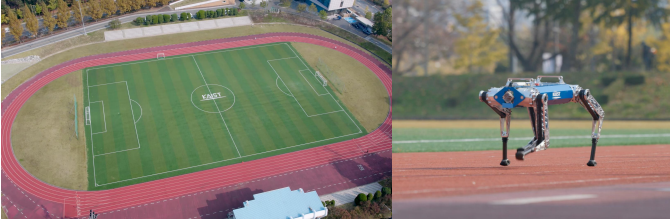


Fig. 10. The presented controller was tested on official athletic track.

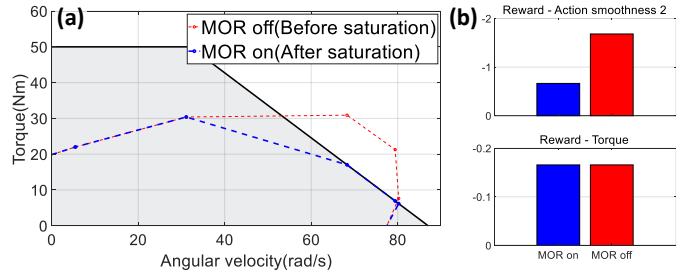


Fig. 11. (a) The red and blue dotted lines that are the  $\tau - \omega$  output of the KFE motor when the robot runs in the simulation at 6.5 m/s subject to the MOR constraint. The red line is the trajectory of the torque command from the network output, and the blue line is the trajectory of the resulting saturated torque command by the MOR constraints. (b) The blue dotted line is more penalized by the action smoothness reward, but the torque applied to the robot is the same as the red trajectory.

the solid line shows the real-world experimental results. The measured velocity and torque of the motor in the experiment are comparable to those in the simulation. The circled region in orange in the KFE in Fig. 8 (c) corresponds to the early stage of the stance phase, where negative energy regeneration occurs. It is observed that the  $\tau - \omega$  trajectory at the region satisfies the MOR constraints by following the slope of the boundary of the MOR constraints in the second quadrant. This result shows that the motor operation well respects MOR constraints which is wider in the 2nd quadrants compared to the 1st quadrants, as described in section II-A.

3) *Outdoor experiments*: We also demonstrated the robot's performance on an official athletics track composed of polyurethane, as shown in Fig. 10. The robot started running with an initial command of 1.2 m/s from rest and accelerated to 5.9 m/s, based on the estimator network, over 6.7 seconds. It took 19.87 seconds to run 100 meters, returning to the same pose having crossed the finish line. The robot is operated with the same command as when it achieved 6.5 m/s on the treadmill but at a slightly lower speed. We suppose that the softness of the urethane track contributes to the drop in maximum speed [4]. A total of nine attempts were made, and the robot completed the 100-meter run without falling over all nine times.

#### D. Discussion

The two colored  $\tau - \omega$  trajectories are shown in Fig. 11. The red line shows the KFE motor output when the robot runs at a speed of 6.5 m/s by the same policy network which is trained without MOR constraints as the red line in Fig. 8. The red trajectory includes torque commands that violate the MOR. The blue line indicates that the torque command in the red line is saturated by the MOR constraint.

The output of the policy network is the desired joint

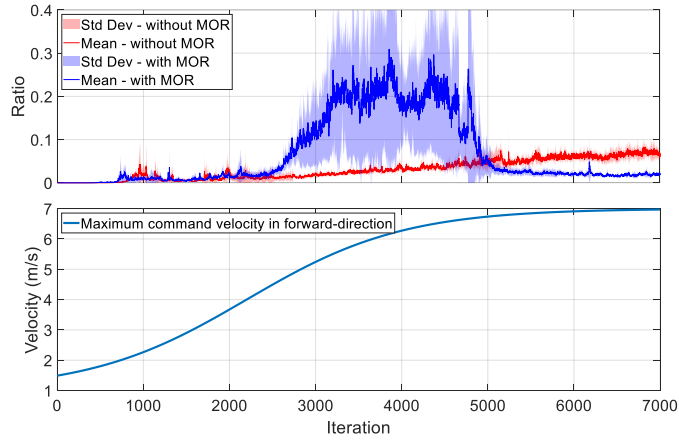


Fig. 12. Upper: Learning curves of different algorithms. The curves show the ratio of the number of torque commands sampled at each iteration that violated the motor operating range. The results shown are obtained from four different random seeds. Lower: Maximum value of the linear velocity command range.

positions, which are converted to joint torques by the PD controller, and the corresponding joint torques are converted to motor torques by the gearbox and applied to the robot. Since the output of the policy network that violates the MOR is clipped to the feasible region by the MOR constraint, the negative reward of the motor torque is the same for both trajectories. However, a sudden action change to produce a torque command that exceeds the MOR is highly penalized by the action smoothness reward. Therefore, the MOR constraint guides the policy network to avoid infeasible torque commands.

It is also observed by the number of data samples that violate the MOR during the learning process shown in Fig. 12. In 400 parallel environments, each episode is 4 seconds long, and the control is done every 0.01 seconds, resulting in 160,000 data samples in every iteration. The vertical axis in Fig. 12 shows the ratio of torque commands exceeding the MOR out of 160,000 data samples. In the early stages of learning low-speed commands that require less motor power, the number of torque commands that violate the MOR is nearly zero, but as the commands increase, the number of data samples that violate the MOR increases. However, the ratio of data samples that violate the MOR in the presence of the MOR constraint converges to a relatively low value compared to learning without the MOR constraint. Therefore, the MOR constraints in the training process help the network to prevent infeasible output commands to the motor. For our study, the robot controller, which was trained without action smoothness reward, demonstrated vibrations when operated on the actual robot.

#### IV. CONCLUSION

This research demonstrates the importance of integrating the robot's physical design elements with reinforcement learning algorithms for high-speed locomotion. Our method incorporates actuator constraints in the training process. We evaluated the proposed method by conducting high-speed running with the designed and built in-house quadruped robot platform, KAIST Hound. The MOR constraints have successfully implemented a 6.5 m/s running test, where the robot accelerated

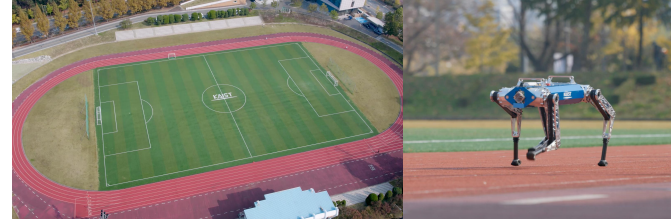


图10. 所提出的控制器在标准田径跑道上进行了测试。

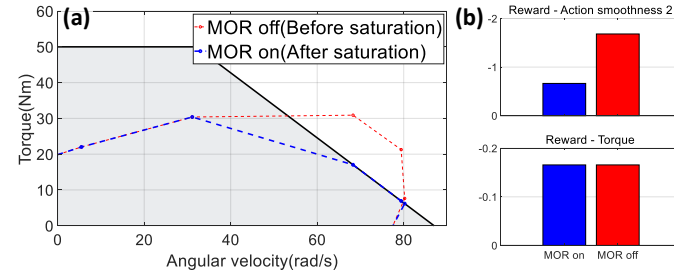


图11. (a) 红色和蓝色虚线是机器人在仿真中以6.5 m/s速度运行并受MOR约束时KFE电机的  $\tau - \omega$  输出。红线是网络输出的扭矩指令轨迹，蓝线是经MOR约束后得到的饱和扭矩指令轨迹。(b) 蓝色虚线受到动作平滑奖励的更大惩罚，但施加到机器人上的转矩与红色轨迹相同。

实线显示现实世界的实验结果。实验中测得的电机速度和转矩与仿真结果相当。图8 (c) 中KFE内橙色圆圈标记的区域对应支撑相的早期阶段，此时发生负能量回收。可以观察到，该区域的  $\tau - \omega$  轨迹通过遵循第二象限中MOR约束边界的斜率来满足MOR约束。这一结果表明，电机运行很好地遵守了MOR约束，与第一象限相比，第二象限中的MOR约束更宽，如第II-A节所述。

3) *室外实验*: 我们还展示了机器人在由聚氨酯铺设的标准田径跑道上的性能，如图10所示。机器人从静止状态以1.2 m/s的初始指令开始奔跑，并基于估计器网络在6.7秒内加速至5.9 m/s。它用时19.87秒跑完100米，并在冲过终点线后恢复到相同的姿态。机器人运行时使用的指令与在跑步机上达到6.5 m/s时相同，但实际速度略低。我们认为聚氨酯跑道的柔软性是导致最高速度下降的原因 [4]。总共进行了九次尝试，机器人九次均未摔倒，全部完成了100米跑。

#### D. 讨论

两条不同颜色的  $\tau - \omega$  轨迹如图11所示。红色线条展示了机器人在未施加MOR约束的情况下，使用与图8中红色线条相同的策略网络，以6.5米/秒速度运行时KFE电机的输出。红色轨迹包含了违反MOR的力矩指令。蓝色线条则表示红色线条中的力矩指令已被MOR约束所饱和和限制。

策略网络的输出是期望的关节

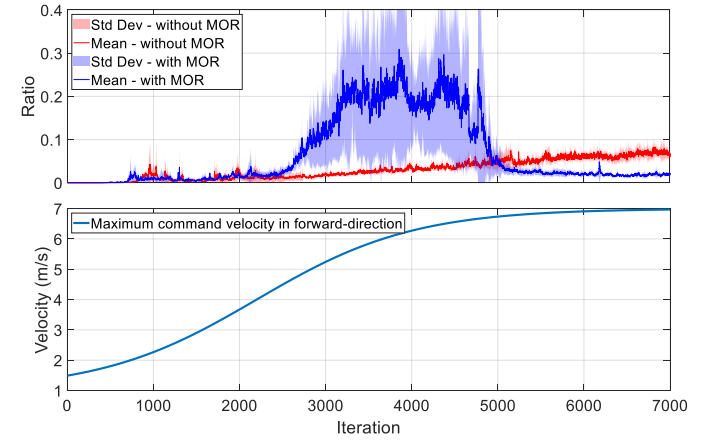


图12. 上图：不同算法的学习曲线。曲线显示了每次迭代中采样的力矩指令违反电机工作范围的比例。所示结果来自四个不同的随机种子。下图：线速度指令范围的最大值。

位置，这些位置通过PD控制器转换为关节转矩，相应的关节转矩通过齿轮箱转换为电机转矩并施加到机器人上。由于违反MOR约束的策略网络输出被裁剪到可行域内，因此两条轨迹的电机转矩负奖励是相同的。然而，为了产生超出MOR的扭矩指令而突然改变动作，会受到动作平滑奖励的严厉惩罚。因此，MOR约束引导策略网络避免不可行的力矩指令。

这一点也可以通过图12所示的学习过程中违反MOR的数据样本数量来观察。在400个并行环境中，每个回合持续4秒，控制每0.01秒执行一次，因此每次迭代产生160,000个数据样本。图12中的纵轴显示了160,000个数据样本中超出MOR的力矩指令比例。在学习初期，由于指令速度较低，所需电机功率较小，违反MOR的力矩指令数量几乎为零；但随着指令速度的增加，违反MOR的数据样本数量也随之增加。然而，在存在MOR约束的情况下，违反MOR的数据样本比例相较于无MOR约束的学习过程收敛到了一个相对较低的值。因此，训练过程中的MOR约束有助于网络防止向电机输出不可行的指令。在我们的研究中，未使用动作平滑奖励训练的机器人控制器在实际机器人上运行时表现出了振动现象。

#### IV. 结论

本研究展示了将机器人物理设计要素与强化学习算法相结合以实现高速运动的重要性。我们的方法在训练过程中融入了执行器约束。我们通过在自行设计并搭建的四足机器人平台KAIST Hound上进行高速奔跑来评估所提方法。电机运行区域约束成功实现了6.5 m/s的奔跑测试，在该测试中机器人从静止状态

from rest to 6.5 m/s in 8 seconds and remained on the treadmill for 35 seconds. By employing MOR constraints from specification sheets, our approach is adaptable to various tasks that require high power, such as jumping and carrying heavy payloads.

In future work, we can compare performance with methodologies that can explicitly incorporate inequality constraints into reinforcement learning [31].

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在8秒内加速至6.5 m/s, 并在跑步机上持续运行了35秒。通过采用规格说明书中的电机运行区域约束, 我们的方法可适用于各种需要高功率的任务, 例如跳跃和搬运重物。

在未来的工作中, 我们可以与能够将不等式约束明确纳入强化学习 [31]的方法论进行性能比较。

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